SU
SINAI UNIVERSITY
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# INT 453: Digital Multimedia

## 2023- 2024

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# Lecture 3

## Digital Multimedia Servers

# Agenda

- Multimedia hardware Requirements
- What is Media Server?
- Simple Model of Distributed Multimedia System
- Types of Multimedia Servers
- Video on demand techniques

# MM hardware requirements

An end system includes two major components; hardware architecture and an operating system.

- The hardware provides raw resources, such as processing power, memory, and data transfer bandwidth.
- The operating system manages these resources so that they can be fully used and the end system as a whole can meet the application's requirements.

# MM hardware requirements

- The hardware should have high processing power and high data transfer throughput, as digital audio and video are very data intensive.
- Many multimedia applications need to access several input/output devices at the same time. So some sort of parallel architecture is preferred.
- The hardware architecture should be scalable to accommodate new input/output devices and applications.

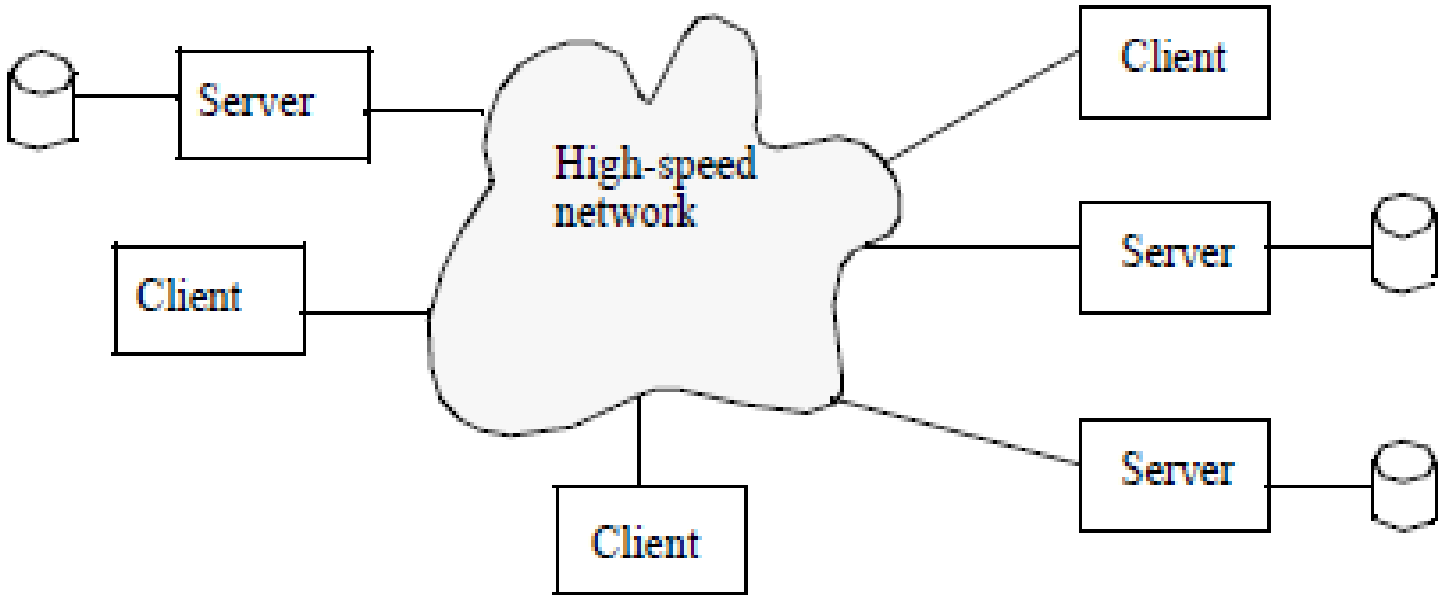
# Multimedia Servers

- It is an end system that can store and manage large amounts of Multimedia information.

- It can serve multiple clients at the same time.

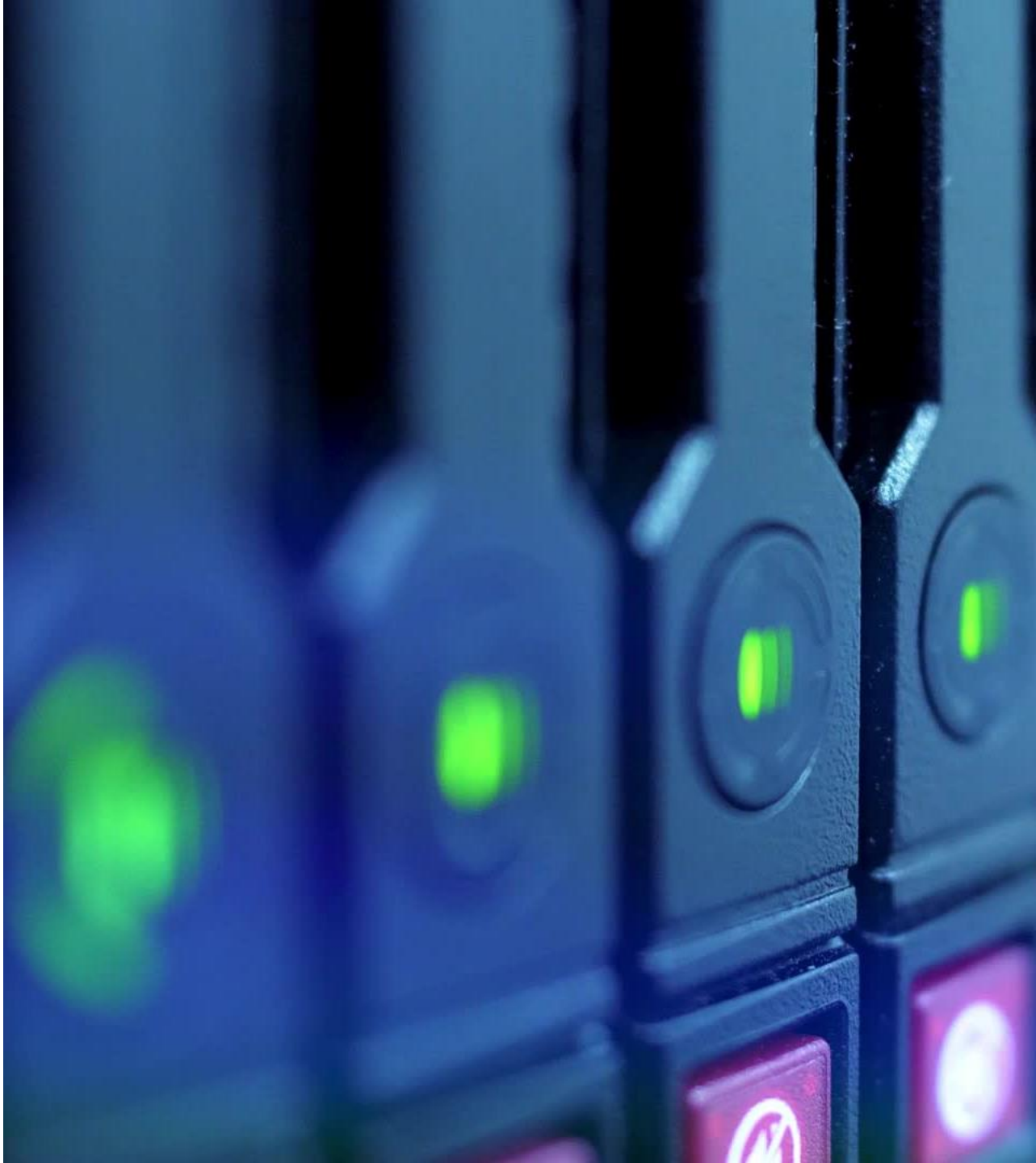
# Simple Model of Distributed Multimedia System

Several clients and servers are interconnected by a high-speed network.



# Types of Media Servers

- Video on Demand (VOD)
- Live streaming

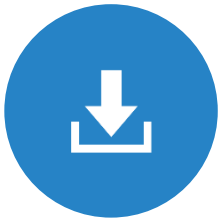




# Types of Media Servers

- The most common use of media servers is probably to deliver **video on demand (VOD),**
- **In VoD,** the media server retrieves prerecorded video content from storage and delivers it across the Internet.
- In the simplest case, the video file is embedded in a web page and stored on the same host as the other components of the page.
- The most common use case for VOD is subscription services like Netflix, Yango Play, Amazon, and Hulu, where media servers located all over the world deliver video to millions of customers from content delivery networks (CDNs).

# VOD Techniques



FILE DOWNLOAD



PROGRESSIVE DOWNLOAD(BITCOMET)



PSUDEO-STREAMING



ADAPTIVE BITRATE STREAMING (WATCH IT)



# VOD Techniques

- **File download**

The video is in a single file, and viewing cannot begin until the entire file has been downloaded from the media server to the player.

- **Progressive download(bitcomet)**

The video is in a single file, but metadata at the beginning of the file enables playback to begin as soon as the metadata and a small amount of video data have been downloaded. The download continues in the background as the video plays.



# VOD Techniques

## Psudeo-streaming

- It can specify the start and end time for the portion of the video it is requesting, and retrieves and delivers ("streams") the corresponding part of the media file.
- This enables the user to fast-forward/reverse or even skip around in the video as it streams.
- Flash and MP4 are popular formats for pseudo-streaming.

## Adaptive bitrate streaming (Watch it)

The media file is segmented into many small files and a playlist is created as a table of contents for the segments, making it even easier for users to skip around in the video.

Popular encodings for adaptive bitrate streaming include HTTP Live Streaming (HLS) from Apple and HTTP Dynamic Streaming (HDS) from Adobe.



# Types of Media Servers

- **Live streaming** is another increasingly popular use of media-streaming servers.
- In this case, the media server delivers content as it is generated in real-time (or with only a slight delay).
- Pay-per-view sporting events are examples of live streaming.
- In contrast to VOD, with live streaming the content provider determines when users view the video (as with broadcast television).
- Live streams can also be recorded for later access as VOD.

# Assignment

- What is the most famous Multimedia servers O.S and how it work ?
- How you create a home multimedia server?

# Review Questions

- Explain the basic requirements of a multimedia operating system.
- Explain the basic media server content.
- Explain the basic video-on-demand techniques.
- Explain Types of Multimedia Servers

# Questions

# Thank You